

CATANDUANES RADAR, Philippines

WMO: 984470

Lat: 13.6298N

Lon: 124.3340E

Elev: 764

StdP: 14.29

Time Zone: 8.00 (E08)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	70.1	71.2	66.2	99.2	71.9	67.3	103.4	72.7	22.9	74.2	18.9	75.2	8.6	40	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	7.1	86.9	80.4	86.0	80.1	85.2	79.7	81.5	85.3	80.8	84.7	80.2	84.0	8.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.3	161.7	84.5	79.6	158.2	83.9	79.1	155.3	83.3	45.9	85.7	45.1	85.1	44.6	84.5	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	18.3	16.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.6	75.5	76.2	77.4	79.0	80.3	80.6	80.0	80.2	80.0	79.3	78.3	76.7
	DBStd	2.54	1.85	1.87	2.04	1.61	1.90	2.01	1.88	1.81	1.93	1.88	1.78	1.97
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10453	791	733	849	869	940	919	931	937	900	909	848	827
	CDD65	4978	326	313	384	419	475	469	466	472	450	444	398	362
	CDH74	38284	1246	1450	2384	3328	4528	4527	4242	4254	3930	3576	2886	1933
CDH80	5513	23	47	163	429	869	951	755	789	682	496	238	71	
Wind	WSAvg	8.7	9.8	9.1	8.8	8.0	7.5	7.4	8.7	8.9	8.9	7.9	9.2	10.1
Precipitation	PrecAvg	149.4	15.2	10.8	9.0	7.8	8.6	10.2	9.6	7.4	9.6	15.9	20.5	24.9
	PrecMax	220.3	41.5	41.5	23.3	36.3	28.6	18.3	18.4	15.6	20.7	35.7	44.3	67.4
	PrecMin	89.5	3.8	1.7	1.1	1.4	2.0	3.2	4.1	2.3	3.3	6.3	3.5	7.2
	PrecStd	42.1	9.3	9.3	6.7	8.5	6.7	3.9	4.0	3.0	4.5	6.7	10.4	14.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.8	82.8	83.9	85.4	87.9	88.2	86.8	87.5	87.7	86.7	84.5	83.5
		MCWB	77.4	77.9	78.2	79.4	80.8	81.2	80.1	80.3	80.6	80.4	79.5	78.9
	2%	DB	80.6	81.6	82.7	84.1	86.1	86.4	85.6	86.1	86.0	85.0	83.3	81.7
		MCWB	76.7	77.1	77.6	78.9	80.1	80.7	79.9	79.8	79.9	79.5	79.0	77.9
	5%	DB	79.6	80.6	81.8	83.3	84.9	85.3	84.7	85.0	84.8	83.9	82.4	80.6
		MCWB	75.9	76.3	77.2	78.5	79.7	80.2	79.5	79.4	79.4	79.0	78.5	77.2
	10%	DB	78.6	79.5	80.8	82.5	83.9	84.3	83.7	83.9	83.6	82.8	81.5	79.6
		MCWB	75.2	75.8	76.7	78.1	79.3	79.8	79.0	79.0	78.9	78.6	78.0	76.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.4	79.0	79.2	81.0	81.9	82.5	81.5	82.2	81.4	81.3	80.6	79.8
		MCDB	81.1	81.9	82.4	83.8	85.9	86.6	85.1	85.7	85.6	84.8	83.3	82.7
	2%	WB	77.3	77.8	78.3	79.8	80.9	81.3	80.6	80.6	80.4	80.1	79.6	78.5
		MCDB	79.7	80.6	81.5	83.0	84.7	85.4	84.6	84.7	84.7	83.5	82.4	80.8
	5%	WB	76.5	77.0	77.8	79.1	80.1	80.6	79.9	79.8	79.8	79.5	79.0	77.7
		MCDB	78.7	79.6	80.8	82.4	83.8	84.6	83.8	83.9	83.9	82.9	81.8	79.8
	10%	WB	75.8	76.3	77.1	78.4	79.6	80.0	79.2	79.1	79.1	78.9	78.3	77.1
		MCDB	77.8	78.7	80.1	81.7	83.2	83.7	83.0	83.2	83.0	82.2	80.9	78.9
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	6.1	7.0	7.0	7.1	6.9	7.5	7.3	7.0	5.8	5.2
		MCDBR	6.7	7.1	7.4	8.0	8.7	8.5	8.4	9.1	9.1	8.6	7.0	6.8
		MCWBR	4.4	4.4	4.3	4.7	4.9	4.8	4.8	5.1	5.0	4.8	4.4	4.3
	5% WB	MCDBR	6.1	6.5	6.8	7.4	7.9	8.3	8.2	8.6	8.6	7.9	6.7	6.3
		MCWBR	4.4	4.5	4.3	4.8	4.8	5.1	5.2	5.4	5.2	5.0	4.5	4.4
Clear-Sky Solar Irradiance	taub	0.449	0.425	0.432	0.416	0.421	0.419	0.422	0.447	0.449	0.441	0.448	0.466	
	taud	2.319	2.403	2.390	2.449	2.461	2.482	2.473	2.382	2.370	2.395	2.354	2.262	
	Ebn at Noon	269	282	281	284	277	275	275	271	272	273	267	260	
	Edh at Noon	40	38	39	37	36	35	35	39	40	38	38	42	
All-Sky Solar Radiation	RadAvg	1222	1544	1725	1972	1929	1761	1562	1624	1605	1453	1348	1151	
	RadStd	145	184	182	127	147	120	148	91	141	90	154	129	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page